ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

# TECHNOLOGY

## PAPER 1 · PRACTICE D · MARKING GUIDELINE

**MARKS: 75**

**Marking guideline for the Academic Central Grade 8 Technology paper 1 · practice d.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

## QUESTION 1

**1** 1.1 A ✓ 1.2 C ✓ 1.3 B ✓ 1.4 D ✓ 1.5 B ✓ 1.6 A ✓ 1.7 B ✓ 1.8 C ✓ 1.9 C ✓ 1.10 A ✓ [10]

## QUESTION 2

**2.1** (a) Frame ✓ (b) Shell ✓ (c) Solid ✓ (3)

**2.2** (a) Tension ✓ (b) Compression ✓ (c) Torsion ✓ (d) Shear ✓ (4)

**2.3** (a) A gusset is a (triangular) plate fixed over a joint ✓; it holds the parts firmly together and stops the joint from bending or moving ✓. (4)
(b) Folding a thin sheet into ridges ✓ makes it much stiffer, so it can carry loads without bending ✓.

**2.4.1** Any sensible brief ✓✓, e.g. "Design and make a stand that holds a cell phone upright on a desk so that a learner can watch videos without holding the phone." (2)

**2.4.2** Any THREE sensible specifications ✓✓✓, e.g. fits phones up to 8 cm wide; holds the phone at an angle of about 60°; made from recycled materials; costs less than R30; does not scratch the phone; leaves space for the charging cable. (3)

**2.4.3** Any TWO ✓✓: a wide base; a low centre of gravity (heavy base); non-slip rubber feet; supports at the back (triangle shape) (2)

**[18]**

## QUESTION 3

**3.1** (a) Resistance to scratching, denting or wear ✓ (b) The ability to carry a load without breaking ✓ (c) Breaks or shatters easily when hit, without bending ✓ (3)

**3.2** Ferrous, any TWO ✓✓: iron; mild steel; stainless steel; cast iron. Non-ferrous, any TWO ✓✓: aluminium; copper; brass; zinc; tin; lead; gold. (4)

**3.3** (a) Iron reacts with oxygen and water (moisture) ✓✓ (4)
(b) Any TWO ✓✓: paint it; galvanise it (zinc coating); oil or grease it; powder-coat it; keep it dry

**3.4** It is the recycling (resin identification) code that shows which type of plastic it is ✓; it helps people and recycling plants to sort the plastics for recycling ✓. (2)

**3.5** The metal pot conducts heat well, so food cooks quickly ✓; plastic is a poor conductor (insulator) of heat, so the handle stays cool and is safe to hold ✓. (2)

**[15]**

## QUESTION 4

**4.1.1** $\text{Gear ratio} = \text{driven teeth} \div \text{driver teeth} = 36 \div 12$ ✓ $= 3:1$ ✓ (2)

**4.1.2** $\text{Speed of C} = 90 \times 12 \div 36$ ✓ $= 30\ \text{r/min}$ ✓ (2)

**4.1.3** Clockwise ✓ (A clockwise, B anticlockwise, C clockwise) (1)

**4.1.4** No ✓ – the speed depends only on the driver and the driven gear; the idler only changes the direction ✓ (2)

**4.1.5** To increase force (torque) ✓, because the driven gear turns more slowly than the driver ✓ (2)

**4.2** Use ✓, e.g. tuning pegs of a guitar, gate openers, conveyors, winches. Advantage ✓, e.g. a very large speed reduction in a small space; it cannot be turned backwards by the load, so it locks in position. (2)

**4.3** Cam (an off-centre or pear-shaped wheel) on a shaft ✓; follower resting on the cam ✓; rotary motion of the cam shown ✓; reciprocating (up-and-down) motion of the follower shown ✓ (4)

**4.4** Lubrication reduces friction between moving parts ✓, which reduces wear and heat, so the machine lasts longer and works more efficiently ✓. (2)

**[17]**

## QUESTION 5

**5.1.1** Battery ✓; two switches in parallel ✓; buzzer in series with the switches ✓; correct symbols and a closed circuit ✓ (4)

**5.1.2** Inputs listed in all four combinations ✓; outputs ✓✓✓ (deduct one for each wrong output): (4)

| Front door (A) | Back door (B) | Buzzer |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

**5.2** Any TWO with a use ✓✓, e.g. push-button switch – doorbell; toggle (light) switch – room lights; slide switch – torch; reed switch – burglar alarm on a window; micro switch – fridge door light (2)

**5.3** $\text{Energy per day} = 2 \times 0{,}5 = 1\ \text{kWh}$ ✓; $\text{for 30 days} = 30\ \text{kWh}$ ✓; $\text{cost} = 30 \times \text{R3,50} = \text{R105}$ ✓ (3)

**5.4** Any TWO ✓✓: never touch switches or plugs with wet hands; do not overload plugs; do not use damaged cords; never push objects into sockets; switch off before unplugging; keep appliances away from water (2)

**[15]**

**TOTAL: 75**

**END OF MARKING GUIDELINE**